Yuming Jin

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RESEARCH INTERESTS

Global Climate Change; Carbon and Oxygen Cycle; Large-scale Atmospheric Dynamics; Carbon-Climate Feedback; Air-Sea interaction

EMPLOYMENT

University of California, Santa Barbara Earth Research Institute	Postdoctoral Fellow	Jan. 2026 -	CA, USA
National Center for Atmospheric Research (NCAR) Earth Observing Laboratory & Climate and Global Dynamics	Advanced Study Program (ASP) Postdoctoral Fellow	Jan. 2024 - Dec. 2025	CO, USA
Scripps Institution of Oceanography, University of California San Diego Geosciences Research Division	Graduate Student Researcher	Sept. 2018 – Dec. 2023	CA, USA

EDUCATION

Scripps Institution of Oceanography, University of California San Diego	Ph.D. Earth Sciences (Climate, Atmospheric Sciences and Physical Oceanography)	2018-2023	CA, USA

Thesis: *Advances in Representing Atmospheric Circulation and Structure for Carbon Cycle Studies* (Awarded UCSD Chancellor's Dissertation Medal)
Advisor: *Dr. Ralph F. Keeling*

Donghua University	B.S. Environmental Sciences	2014-2018	Shanghai,

China

Graduation with honor: College Graduate Excellence Award of Shanghai

Thesis: *Preparation and application of Bi_2MoO_6-based recyclable visible-light driven photocatalyst*

(Awarded University Best Undergraduate Thesis)

Advisor: *Dr. Lisha Zhang*

LIST OF PUBLICATIONS [Google Scholar; ResearchGate]

MANUSCRIPTS UNDER REVIEW or IN PREPARATION

14. **Jin, Y.**, Morgan, E, Stephens, B., Landschützer, P., DeVries, T.: Ocean heat-oxygen-carbon coupling constrains the climate-driven carbon sink, **Under review at Global Biogeochem. Cy.**
13. Morgan, E., **Jin, Y.**, Tohjima, Y., Keeling, R.: Atmospheric O_2 measurements show a strengthening ocean carbon sink and a stagnate terrestrial carbon sink since 1990, **Under review at Science**
12. **Jin, Y.,** Stephens, B., Morgan, E., Manizza, M., Sharp, J.: Global Gridded Air-sea O_2 Flux Inferred from Machine Learning-based Dissolved Oxygen Products, **Under review at ESSD**

PEER-REVIEWED PUBLICATIONS

11. Stephens, B., **Jin, Y.**, Sweeney, C., McKain, K., Gaubert, B., Baker, D., Basu, S., Bertolacci, M., Chevallier, F., Commane, R., Crowell, S., Deng, F., Johnson, M., Keeling, R., Liu, J., Maity, S., Morgan, E., Patra, P., Philip, S., Wofsy, S., Zammit-Mangio, A., Global airborne surveys reduce uncertainty in latitudinal carbon budgets and suggest biases in process-based models, ***PNAS***, 2026, 125(25), e2523984123, https://doi.org/10.1073/pnas.2523984123

    Highlights: NCAR News
10. **Jin, Y.,** Stephens, B., Long, M., Manizza M., Lovenduski, N., Nevison, C., Morgan, E., Keeling, R.: Atmospheric Oxygen Constraints on Southern Ocean Productivity and Drivers of Carbon Uptake, ***Nat. Geosci.,*** 2026, 4, https://doi.org/10.1038/s41561-026-01944-z

    Highlights: Nature Geoscience News & Views, NCAR News, earth.com
9. **Jin, Y.**, Stephen, B., Long, M., Chandra, N., Chevallier, F., Hooghiem, J., Luijkx, I., Maksyutov, S., Morgan, E., Niwa, Y., Patra, P., Rödenbeck, C., Vance J.: The Atmospheric Potential Oxygen forward Model Intercomparison Project (APO-MIP): Evaluating simulated atmospheric transport of air-sea gas exchange tracers and APO flux products, ***Geosci. Model Dev.***, 2025, 18, 5937–5969, https://doi.org/10.5194/gmd-18-5937-2025
8. Liu, Z., Rogers, B., Keppel-Aleks, G., Helbig, M., Ballantyne, A., Kimball, J., Chatterjee, A., Foster, A., Kaushik, A., Virkkala, A., Burrell, A., Schwalm, C., Sweeney, C., Schuur, E., Dean, J., Watts, J., Kim, J., Wang, J., Hu, L., Welp, L., Berner, L., Mauritz, M., Farina, M., Mack, M., Parazoo, N., Madani, N., Keeling, R., Commane, R., Goetz, S., Piao, S., Natali, S., Wang, W., Buermann, W., Walker, X., Lin,

X., Wang, X., **Jin, Y.**, Yu, K., Zhang, Y.: Drivers and trends of enhanced seasonal CO_2 amplitude in northern high latitudes, ***Nat. Rev. Earth & Env.***, 2024, https://doi.org/10.1038/s43017-024-00600-7

7. **Jin, Y.**, Keeling, R. F., Long, M. C., Stephens, B. B., Patra, P., Rödenbeck, C., Morgan, E. J., Kort, E. A., Sweeney, C.: Improved Atmospheric Constraint of the Southern Ocean CO_2 Exchange, ***PNAS***, 2024, 121(6), e2309333121, https://doi.org/10.1073/pnas.2309333121
6. **Jin, Y.**, Stephens, B. B, Keeling, R. F., Morgan, E. J., Rödenbeck, C., Patra, P., Long, M. C.: Seasonal Tropospheric Distribution and Air-sea Fluxes of Atmospheric Potential Oxygen from Global Airborne Observations, ***Global Biogeochem. Cy.***, 2023, 37, e2023GB007827, https://doi.org/10.1029/2023GB007827
5. **Jin, Y.**, Keeling, R. F., Rödenbeck, C., Patra, P., Piper, S., and Schwarzman, A.: Impact of Changing Winds on the Mauna Loa CO_2 Seasonal Cycle in Relation to the Pacific Decadal Oscillation, ***JGR-Atmosphere***, 2022, 127(13), https://doi.org/10.1029/2021JD035892
4. Morgan, E., Manizza, Manfredi., Keeling, R. F., Resplandy, L., Mikaloff-Fletcher, S.E., Nevison, C.D., **Jin, Y.**, Bent, J.D., Aumont, O., Doney, S.C., Dunne, J.P., John, J., Lima, I.D., Long, M.C., Rodgers, K.B.: An atmospheric constraint on the seasonal air–sea exchange of oxygen and heat in the extratropics, ***JGR-Ocean***, 2021, 126(8), e2021JC017510, https://doi.org/10.1029/2021JC017510
3. **Jin, Y.**, Keeling, R. F., Morgan, E. J., Ray, E., Parazoo, N. C., and Stephens, B. B.: A mass-weighted isentropic coordinate for mapping chemical tracers and computing atmospheric inventories, ***Atmos. Chem. Phys.***, 2021, 21, 217–238, https://doi.org/10.5194/acp-21-217-2021

Prior to graduate training

2. **Jin, Y.**, Shen, X., Liu, Z., Wang, Z., Zhu, B., Xu, P., Luo, L., Zhang, L.: Synthesis of $NiTiO_3$–Bi_2MoO_6 core–shell fiber-shaped heterojunctions as efficient and easily recyclable photocatalysts. ***New J. Chem.*** 2018; 42(1): 411-9, https://doi.org/10.1039/C7NJ03367B
1. Lou, X., Fang, C., Geng, Z., **Jin, Y.**, Xiao, D., Wang, Z., Liu, J., Guo, Y.: Significantly enhanced base activation of peroxymonosulfate by polyphosphates: Kinetics and mechanism. ***Chemosphere***. 2017; 173: 529-34, https://doi.org/10.1016/j.chemosphere.2017.01.093

OTHER PUBLICATIONS

1. Betts, R., Jones, C., **Jin, Y.**, Keeling, R. F., Kennedy, J. J., Knight, J. R.: Analysis: What impact will the coronavirus pandemic have on atmospheric CO_2, Carbon Brief, 2020

SEMINAR (INVITED) TALKS

7. Atmospheric Constraints on Southern Ocean Carbon Cycle, UCSB IGPMS Seminar, 21 April, 2026
6. Constraining Southern Ocean Seasonal CO_2 Uptake with Atmospheric O_2 and CO_2 Measurements, SOFLUX Webinar, 6 June, 2024
5. Improving estimates of carbon sources, sinks, and atmospheric transport using global airborne data, JPL Carbon Club Seminar, 18 April, 2024
4. Using airborne CO_2 observations and a mass-indexed isentropic coordinate to test transport models and improve surface flux estimates, OCO2-Flux telecon, 23 January, 2024
3. Quantification of seasonal air-sea CO_2 and O_2 exchange at regional to global scale using data from airborne campaigns, Shanghai Jiao Tong University, School of Oceanography Seminar, 6 April, 2023
2. Atmospheric CO_2 and O_2 observations as indicators of global climate and biogeochemistry change, Carbon Monitor Seminar, 24 July, 2022
1. Improved Atmospheric Constraint of the Southern Ocean CO_2 Sink, BGC-Argo Science Meeting, 18 May, 2022

CONFERENCE and WORKSHOP PRESENTATIONS

16. **Jin Y.**: Atmospheric O_2 Constraints on Ocean Biogeochemistry and Physics, CLIVAR Oxygen Workshop, Beijing, 23-26 Oct., 2025 (Oral)
15. **Jin Y.**: Atmospheric constraints on the Southern Ocean productivity and seasonal air-sea CO_2 and O_2 exchange, CLIVAR Summit 2025, Boulder, 21-23 July, 2025 (Oral)
14. **Jin Y.**: An Isentrope-Based Framework for Diagnosing Tracer Transport and Surface Fluxes, TransCom Workshop, Online, 28 May, 2025 (Oral)
13. **Jin, Y.**: Improving estimates of carbon sources, sinks, and atmospheric transport using global airborne data and a transformed isentrope coordinate, ICDC 2024, Manaus, 29 July - 2 Aug., 2024 (Oral)
12. **Jin, Y.**: Refining transport models: leveraging airborne tracer observations and cross-isentrope diabatic mixing metrics, TransCom-24, Boulder, 28 May, 2024 (Oral)
11. **Jin, Y.**: Southern Ocean Seasonal O_2 and CO_2 Exchange Inferred from Airborne Observations, Ocean Sciences Meeting 2024, New Orleans, 18-23 Feb., 2024 (Poster)
10. **Jin, Y.** et al.: Long-term and decadal changes of Mauna Loa CO_2 seasonal cycle in relation to wind shifts driven by Pacific Decadal Oscillation, AGU Fall Meeting 2023, San Francisco, 11-15 Dec., 2023 (Poster)
9. **Jin, Y.** et al.: Seasonal Air-sea Atmospheric Potential Oxygen Flux Inferred from Global Airborne Observations, 4th Workshop on Atmospheric Oxygen, Brunswick, 23-25 Aug., 2023 (Oral)
8. **Jin, Y.** et al.: Using airborne CO_2 observations and a mass-indexed isentropic coordinate to test transport models and improve surface flux estimates, NOAA GML Global Monitoring Annual Conference 2023, Boulder, 23-24 May, 2023 (Poster)

7. **Jin, Y.** et al.: Constraining Atmospheric Transport Models and Improving Surface CO_2 Flux Estimates Using Airborne CO_2 Observations and a Mass-Indexed Isentropic Coordinate, AMS 2023, Denver, CO, 9 Jan. – 12 Jan., 2023 (Oral)
6. **Jin, Y.** et al.: Using airborne CO_2 observations and a mass-indexed isentropic coordinate to test transport models and improve surface flux estimates, TransCom Meeting, Wageningen, Netherlands, 16-17 Sept., 2022 (Oral)
5. **Jin, Y.** et al.: Constraining the Seasonal Cycle of the Southern Ocean CO_2 Uptake with Airborne Observations, Ocean Sciences Meeting 2022, Virtual Meeting, 28 Feb. – 4 Mar., 2022 (Oral)
4. **Jin, Y.** et al.: Constraining the Seasonal Cycle of the Southern Ocean CO_2 Uptake with Airborne Observations, AGU Fall Meeting 2021, New Orleans, 13-18 Dec., 2021 (Oral)
3. **Jin, Y.** et al.: Mapping Atmospheric CO_2 and O_2 from Airborne Observations Using a Mass-weighted Isentropic Coordinate, NOAA GML Virtual Global Monitoring Annual Conference 2021, Virtual Meeting, 24-28 May, 2021 (Poster)
2. **Jin, Y.** et al.: Mapping the CO_2 Seasonal Cycle from Airborne Observations Using a Mass-Weighted Isentropic Coordinate, American Meteorology Society Annual Meeting, Virtual Meeting, 9-15 Jan., 2021 (Oral)
1. **Jin, Y.** et al.: Precise quantification of seasonal air-sea exchanges of O_2 at the hemispheric scale using data from the ATom, ORCAS, and HIPPO airborne campaigns, Ocean Sciences Meeting 2020, San Diego, CA, 16-21 Feb., 2020 (Poster)

TEACHING EXPERIENCE

Teaching Assistant: SIO 117: Physical Basis of Global Warming (2023)

Tutor: Math Bootcamp (2019)

HONORS / AWARDS

UCSD Chancellor’s Dissertation Medal	2024
NCAR ASP Postdoc Fellowship	2023
SIO Department Graduate Student Travel / Research Award	2023
Henry & Grace Doherty Fellowship Fund	2018
Hariko Quirk Endowment Fund	2018
National Scholarship (Ministry of Education of China)	2017
First Class Scholarship & Outstanding Student Award (Donghua University)	2015 & 2016 & 2017

SERVICE & OUTREACH

Reviewer: AGU Advances, Atmospheric Chemistry and Physics, Biogeosciences, Earth and Space Science, Earth System Science Data, Environmental Research Communications, Environmental Research Letters, Global Biogeochemical Cycles, Journal of Advances in Modeling Earth Systems, Journal of Geophysical Research-Atmosphere, Journal of Geophysical Research-Ocean, Nature Communication, Science Advances, Scientific Report

Committee Member:

ASP Seminar Series Committee (2023-2025)

AI4Carbon Committee Organizer (2024-present)

Judge: Scripps Student Symposium (2021)

Conference Organizer/Convener:

NCAR Distinguished Lecture Series, 16-17 April 2024, Boulder, USA

NCAR Expanding Horizon Seminar series, 19 September 2024, Boulder, USA

EGU2025: Understanding feedbacks between greenhouse gas exchange processes and climate variability using in situ observations, remote sensing, and machine learning, 27 April – 2 May 2025, Vienna, Austria

WORKSHOPS AND TRAININGS

CESM Tutorial (Aug. 2020)

Summer School for Inverse Modeling of Greenhouse Gases Summer School (June 2024)

Workshop for New Advances in Land Carbon Cycle Modeling (June 2026)

GO-BGC Workshop (Aug. 2026)

TECHNICAL SKILLS:

Programming Language: R (expert), Python (intermediate), Fortran (intermediate), Julia (advanced beginner), Bash

Modeling Experience: CESM, TM3, HYSPLIT, Jena CarboScope Inversion